

Customer Shopping Behavior Analysis

1. Project Overview

This project focuses on analyzing customer shopping behavior using retail transaction data. The objective is to understand customer purchasing habits, spending patterns, product preferences, and subscription trends. The project also aims to generate useful business insights that can help improve customer experience and support better business decisions. Python was used for data preprocessing, MySQL for data analysis, and Power BI for creating an interactive dashboard.

2. Dataset Summary

Rows: 3,900

Columns: 18

Key Features

- Customer information: Age, Gender, Location, Subscription Status
- Product information: Item Purchased, Category, Purchase Amount, Size, Color, Season
- Shopping details: Discount Applied, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type

Missing Values

The dataset contains missing values in the Review Rating column, which were handled during the data cleaning process.

3. Exploratory Data Analysis Using Python

The dataset was prepared and cleaned in Python before performing the analysis.

- **Data Loading:** Imported the dataset using the Pandas library.
- **Initial Data Exploration:** Examined the dataset using `df.info()`, `df.head()`, and `df.describe()` to understand the structure, data types, and summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN

Previous Purchases	Payment Method	Frequency of Purchases
3900.000000	3900	3900
NaN	6	7
NaN	PayPal	Every 3 Months
NaN	677	584
25.351538	NaN	NaN
14.447125	NaN	NaN
1.000000	NaN	NaN
13.000000	NaN	NaN
25.000000	NaN	NaN
38.000000	NaN	NaN
50.000000	NaN	NaN

Missing Value Treatment: Identified missing values using `df.isnull().sum()` and filled the missing Review Rating values with the median rating for each product category.

```

Customer ID          0
Age                  0
Gender               0
Item Purchased       0
Category             0
Purchase Amount (USD) 0
Location            0
Size                0
Color               0
Season              0
Review Rating        37
Subscription Status  0
Shipping Type        0
Discount Applied     0
Promo Code Used      0
Previous Purchases   0
Payment Method       0
Frequency of Purchases 0
dtype: int64

```

Column Standardization: Renamed the column names to snake_case to improve readability and maintain consistent naming.

Feature Engineering:

Created an Age Group column by grouping customers based on age.

Created a purchase frequency field to simplify customer behavior analysis.

Data Validation: Verified data consistency and removed unnecessary columns after checking for duplicate information.

Database Integration: Exported the cleaned dataset into MySQL for SQL-based business analysis.

4. Data Analysis Using MySQL (Business Transactions)

After cleaning the dataset in Python, the data was imported into MySQL for business analysis. SQL queries were used to answer important business questions and identify customer purchasing trends. The analysis helped understand customer behavior, product performance, and revenue generation.

The following SQL queries were performed:

1. Revenue by Gender

Calculated the total purchase amount generated by male and female customers. This analysis helped identify which gender contributed more to the overall revenue.

gender	revenue
Male	157890
Female	75191

2. High-Spending Customers Using Discounts

Identified customers who used discounts but still spent more than the average purchase amount. This helped understand the purchasing behavior of high-value customers.

customer id	purchase amount (usd)
2	64
3	73
4	90
7	85
9	97
12	68
13	72

3. Top 5 Products by Review Rating

Calculated the average review rating for each product and displayed the top five highest-rated products based on customer feedback.

item purchased	avg_rating
Gloves	3.8627737226277383
Sandals	3.8446540880503144
Boots	3.818881118881119
Hat	3.801307189542483
Skirt	3.7853503184713366

4. Shipping Type Comparison

Compared the average purchase amount between customers who selected Standard Shipping and Express Shipping to understand the impact of shipping methods on customer spending.

shipping type	avg_purchase
Express	60.4752
Free Shipping	60.4104
Next Day Air	58.6312
Standard	58.4602
2-Day Shipping	60.7337
Store Pickup	59.8938

5. Subscribers vs Non-Subscribers

Compared the average purchase amount and total revenue generated by subscribers and non-subscribers to evaluate the value of subscription services.

subscription status	average_spend	total_revenue
Yes	59.4919	62645
No	59.8651	170436

6. Products with the Highest Discount Usage

Identified the top five products that received discounts most frequently. This analysis helped understand which products relied more on promotional offers.

item purchased	total_purchases	discount_purchases	discount_percentage
Hat	154	77	50.00
Sneakers	145	72	49.66
Coat	161	79	49.07
Sweater	164	79	48.17
Pants	171	81	47.37

7. Customer Segmentation

Classified customers into New, Returning, and Loyal groups based on their previous purchase history to better understand customer retention.

customer_segment	customer_count
Loyal	3116
Returning	784

8. Top 3 Products in Each Category

Displayed the three most frequently purchased products within each product category using SQL ranking functions.

item_rank	category	item purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145

9. Repeat Buyers and Subscription Analysis

Analyzed whether customers with more previous purchases were more likely to subscribe to the membership program.

subscription status	repeat_buyers
Yes	958
No	2518

10. Revenue by Age Group

Calculated the total revenue generated by different age groups to identify the most valuable customer segments.

age_group	total_revenue
56+	65256
46-55	45619
26-35	44342
36-45	43234
18-25	34630

5. Power BI Dashboard

After completing the data cleaning in Python and analysis in MySQL, an interactive dashboard was created using Power BI. The dashboard was designed to present the analysis in a simple and visual format, making it easier to understand customer shopping behavior and business performance.

The dashboard includes the following visualizations:

KPI Cards: Display Total Revenue, Total Customers, Average Purchase Amount, and Total Orders.

Bar Charts: Compare revenue by gender, product category, and age group.

Pie Chart: Show the distribution of subscribers and non-subscribers.

Column Chart: Compare purchase amounts based on shipping type.

Line Chart: Display purchase trends across different seasons.

Slicers: Allow users to filter the dashboard by Gender, Category, Season, Subscription Status, and Age Group.

The dashboard provides an interactive way to explore the data and helps users identify important business trends, customer preferences, and purchasing patterns. It supports better decision-making by presenting key insights in a clear and user-friendly format.



6. Business Recommendations

- Increase customer subscriptions by offering exclusive discounts and loyalty rewards.
- Focus marketing campaigns on the customer groups that generate the highest revenue.
- Promote top-rated and best-selling products to increase customer engagement and sales.
- Review the discount strategy to ensure discounts improve sales without reducing overall profit.
- Encourage repeat purchases by introducing reward programs and personalized offers for loyal customers.
- Improve the shopping experience by providing faster shipping options and better customer service.

7. Conclusion

This project successfully analyzed customer shopping behavior using Python, MySQL, and Power BI. The analysis provided valuable insights into

customer spending patterns, product performance, subscription behavior, and purchasing trends. The interactive dashboard makes it easier to understand the data and supports better business decision-making. Overall, this project demonstrates how data analytics can help businesses improve customer satisfaction and increase sales.